HiStar System Commands:

Each power supply channel can be remotely controlled via an Ethernet link using a predefined set of commands. To ensure a correct communication with each module some rules need to be followed:

- commands to the power supply must be sent with a '**\r**' (*carriage return*, 0x0D) termination character;
- replies from the power supply controller have a '**\r**' (*carriage return*, 0x0D) termination character;

Command:	Description:	Type:
MRI	Ouput current readback (A)	R
MRV	Ouput voltage readback (V)	R
MRW	Output active power readback (W)	R
MWI	Output current set (A)	W
MRF	EEPROM cell field read	R
MWF	EEPROM cell field write	W
MRG	EEPROM cell value read	R
MWG	EEPROM cell value write	W
MRH	Ouput current readback (hexadecimal)	R
MWH	Ouput current set (hexadecimal)	W
MRM	Output current set using a ramp (A)	W
MRT	Output transistors temperature read (° C)	R
RTS	Shunt resistor temperature read (° C)	R
MRP	DC-Link voltage read (V)	R
MON	Enable channel output	W
MOFF	Disable channel output	W
MRID	Read channel identification	R
MST	Status register read	R
MRESET	Status register reset	W
PASSWORD	Password to unlock protected EEPROM cells	W
MGC	Ground leakage current read (A)	R
MGLST	Global list read	R
MUP	Update channel parameters reading the EEPROM	W
PTP	Update PID digital regulator parameters	W
SIP	Set channel IP number	W
MAC	Channel MAC address read	R
BON	Set Bulk ON	W
BOFF	Set Bulk OFF	W
VER	Control firmware version read	R

Table 1: List of commands

The power supply controller replies every time a termination character '**\r**' is received and it could have different behaviors:

- an acknowledgment '**#AK\r**' string is sent back in case of a correct setting command;
- a non-acknowledgment '**#NAK\r**' string is sent back in case of a wrong/unrecognized command or if the system is in ***local*** operation mode and a ***write*** command is sent to the controller (write commands are marked in the Type column of Table 1 with a W);
- a standard reply, preceded by a '**#**' and followed by a '**\r**' character, is sent back as response to a reading command.

A brief description for each command is herein presented.

'MRI' Command

The **'MRI\r'** command returns the value of the power supply actual current output. Replies from the power supply A2630BS controller are in the form '**#MRI:*value*\r**' where ***value*** is the current value (in A=Ampere).

'MRV' Command

The **'MRV\r'** command returns the value of the power supply actual voltage at the output terminals. Replies from the power supply A2630BS controller are in the form '**#MRV:*value*\r**' where ***value*** is the voltage value (in V=Volt).

'MRW' Command

The **'MRW\r'** command returns the value of the estimated active power supplied to the load by the power supply. Replies from the power supply A2630BS controller are in the form '**#MRW:*value*\r**' where ***value*** is the active power value (in W=Watt).

'MWI' Command

The **'MWI'** command is used to set the value of the desired output current set-point: the correct form is '**MWI:*value*\r**' where ***value*** is the current value (in A=Ampere). The A2630BS controller will reply with an acknowledgment command '**#AK\r**' if the value is correctly set and with a '**#NAK\r**' if:

- the set value is out-of-range (maximum settable current value is user-defined and stored in the channel EEPROM),
- the channel is OFF (it is necessary to turn the module on with the 'MON' command first),
- the controller is still performing a ramp (see command 'MRM'),
- the system is in local mode.

'MRF' Command

The **'MRF'** command returns the value stored in the "field" parameter of a desired EEPROM cell: correct form is '**MRF:*cell*\r**' where ***cell*** is the desired EEPROM cell number. Replies from the power supply A2630BS controller are in the form '***value*\r**' where ***value*** is an ASCII value.

‘MWF’ Command

The **‘MWF’** command lets users write a desired “field” value in a defined EEPROM cell: correct form is ‘**MWF:*cell:value*\r**’ where ***cell*** is the desired EEPROM cell number and ***value*** is an ASCII string. The A2630BS controller will reply with an acknowledgment command ‘**#AK\r**’ if the value is correctly set and with a ‘**#NAK\r**’ if:

- command is incorrectly fed,
- the selected cell is password protected (see PASSWORD command),
- the system is in local mode.

‘MRG’ Command

The **‘MRG’** command returns the value stored in the “value” parameter of a desired EEPROM cell: correct form is ‘**MRG:*cell*\r**’ where ***cell*** is the desired EEPROM cell number. Replies from the power supply A2630BS controller are in the form ‘***value*\r**’ where ***value*** is an ASCII value.

‘MWG’ Command

The **‘MWG’** command lets users write a desired “value” value in a defined EEPROM cell: correct form is ‘**MWG:*cell:value*\r**’ where ***cell*** is the desired EEPROM cell number and ***value*** is an ASCII string. The A2630BS controller will reply with an acknowledgment command ‘**#AK\r**’ if the value is correctly set and with a ‘**#NAK\r**’ if:

- command is incorrectly fed,
- the selected cell is password protected (see PASSWORD command),
- the system is in local mode.

‘MRH’ Command

The **‘MRH\r’** command is for debugging purposes only and it returns the value of the power supply actual current output in a hexadecimal notation. Replies from the power supply A2630BS controller are in the form ‘**#MRH:*value*\r**’ where ***value*** is the ADC raw current value (in hexadecimal code).

‘MWH’ Command

The **‘MWH’** command is used to set the value of the desired output current set-point: the correct form is ‘**MWH:*value*\r**’ where ***value*** is the ADC raw current value in hexadecimal notation. The A2630BS controller will reply with an acknowledgment command ‘**#AK\r**’ if the value is correctly set and with a ‘**#NAK\r**’ if:

- the sent command is incorrect,
- the channel is OFF (it is necessary to turn the module on with the 'MON' command first),
- the controller is still performing a ramp (see command 'MRM'),
- the system is in local mode.

‘MRM’ Command

The **‘MRM’** command is used to set the value of the desired output current set-point: the correct form is ‘**MRM:*value*\r**’ where ***value*** is the current value (in A=Ampere). The difference between the ‘MWI\r’ command and the ‘MRM\r’ command is that the first one generates a direct change in output current characterized by the PID engine parameters (no slew-rate defined, ideally suited for small changes and feedback purposes) while the second one goes from the previous to the actual current value performing a ramp, defined by a slew-rate (in A/s) stored in the EEPROM. The A2630BS controller will reply with an acknowledgment command ‘**#AK\r**’ if the value is correctly set and with a ‘**#NAK\r**’ if:

- the set value is incorrect.
- the channel is in OFF,
- the channel is still ramping (it is necessary to wait for the power supply to end the previous ramp),
- the system is in local mode.

‘MRT’ Command

The **‘MRT\r’** command returns the value of the temperature directly measured on the output stage MOSFET heatsink. Replies from the power supply A2630BS controller are in the form ‘**#MRT:*value*\r**’ where ***value*** is the temperature value (in °C=Celsius). Even if the internal ADCs have a 16-bit resolution, this value is presented to the user with a 0.1 °C (= 0.1 K) resolution.

‘MRTS’ Command

The **‘MRTS\r’** command returns the value of the temperature directly measured on the shunt heatsink. Replies from the power supply A2630BS controller are in the form ‘**#MRTS:*value*\r**’ where ***value*** is the temperature value (in °C=Celsius). Even if the internal ADCs have a 16-bit resolution, this value is presented to the user with a 0.1 °C (= 0.1 K) resolution.

‘MRP’ Command

The **‘MRP\r’** command returns the value of the actual DC-Link voltage. Replies from the power supply A2630BS controller are in the form ‘**#MRP:*value*\r**’ where ***value*** is the DC-Link voltage value (in V=Volt). Even if the internal ADCs have a 16-bit resolution, this value is presented to the user with a 100 mV resolution.

‘MON’ Command

The **‘MON\r’** enables the power supply output driver. The power supply automatically sets current to 0A (zero) before enabling the output. The A2630BS controller will reply with an acknowledgment command ‘**#AK\r**’ if the command is correctly executed and with a ‘**#NAK\r**’ if:

- the channel is in fault (it is necessary to reset the status register after a generic fault condition in order to turn on the power supply again - see command MRESET),
- bulk is off (see command BON to turn on bulk),
- the system is in local mode.

'MOFF' Command

The **'MOFF\r'** disables the power supply output driver; the power supply itself sets current to 0A (zero) with a 20 A/s slew-rate before disabling the output, this is done to avoid voltage overshoots (that would be anyway smoothed and limited by the crowbar and over-voltage clamping protections) especially for high currents and strongly inductive loads. Replies from the power supply A2630BS controller are in the form '**#AK\r**' or '**#NAK\r**' if:

- the channel is ramping,
- the system is in local mode.

'MRID' Command

The **'MRID\r'** command returns the value of the module identification name (example: "SkewMag1.1"). The reply from the power supply A2630BS controller contains the value stored in cell 27 of the module EEPROM and it assumes the form '***name*\r**' where ***name*** is an ASCII string. This command is equivalent to the 'MRG:27\r' command.

'MST' Command

The **'MST\r'** command returns the value of the power supply status register (32 bit). Replies from the power supply A2630BS controller are in the form '**#MST:*value*\r**' where ***value*** is the ASCII representation of the register value (8 hexadecimal digits, corresponding to the 32bit wide status register).

Status Register Structure (32-bit)	
Bit [31:28]	*reserved*
Bit 27	BULK REDUNDANCY
Bit 26	AUX EARTH OK
Bit 25	*reserved*
Bit 24	BULK ON
Bit [23:16]	EXTERNAL INTERLOCKS[7:0]
Bit [15:12]	*reserved*
Bit 11	REGULATOR FAULT
Bit 10	GND CURRENT
Bit 9	DC UNDERVOLTAGE
Bit 8	SHUNT TEMP
Bit 7	MOSFET TEMP
Bit 6	CROWBAR
Bit 5	INPUT OVERCURRENT
Bit 4	DSP ERROR
Bit 3	LOCAL
Bit 2	WARNING
Bit 1	FAULT
Bit 0	MODULE ON

Table 2: Status Register

‘MRESET’ Command

The **‘MRESET\r’** command performs a reset of the module status register: this is needed, for example, to enable the channel output again after a fault condition has been fixed. Reply from the A2630BS controller is ‘**#AK\r**’.

‘PASSWORD’ Command

Password command has the form of ‘**PASSWORD:*value*\r**’, where ***value*** is the ASCII string that allows users to write to some EEPROM protected cells. These cells include calibration parameters, calibration date, serial number, etc. and the password is not given to the end-user of the power supply since some parameters as module identification, PID parameters, slew-rate value, etc. are not password protected and can be easily modified. The protected cells are marked in Table 3.

‘MGC’ Command

The **‘MGC\r’** command returns the value of the actual ground leakage current. Replies from the power supply A2630BS controller are in the form ‘**#MGC:*value*\r**’ where ***value*** is the ground leakage current (in A=Ampere).

‘MGLST’ Command

The **‘MGLST\r’** command returns a list of power supply values, which are:

- the value of actual current output,
- the value of the voltage at the output terminals,
- the value of the status register,
- the value of the actual ground leakage current,
- the value of the last output current set-point.

Replies from the power supply A2630BS controller are represented in the following form: ‘**#MGLST:*value_i,value_v,value_s,value_gnd,value_sp*\r**’ where ***value_i*** is the current value (in A=Ampere), ***value_v*** is the voltage value (in V=Volt), ***value_s*** is the ASCII representation of the register value (8 hexadecimal digits, corresponding to the 32-bit register), ***value_gnd*** is the ground leakage current (in A=Ampere) and ***value_sp*** is the last current set-point value (in A=Ampere).

‘MUP’ Command

The **‘MUP\r’** command performs an update of the power supply A2630BS parameters with the parameters read from the EEPROM; as an example, the command MWF updates only the content of the selected EEPROM field but not the corresponding A2630BS controller parameter. In order for the A2630 controller to update its parameter it is necessary to perform the ‘MUP\r’ command. Replies from the power supply A2630BS controller are in the form ‘**#AK\r**’ or ‘**#NAK\r**’ if the system is in local mode.

‘PTP’ Command

The **‘PTP\r’** command performs an update of the power supply A2630BS PID controller regulation parameters. This command reads the updated parameters with command MUP first and then sends it to the PID controller. For example: to modify the PID parameters it is necessary to modify the content of the EEPROM fields (Table 3) and then to execute the commands MUP and PTP. Replies from the power supply A2630BS controller are in the form ‘**#AK\r**’ or ‘**#NAK\r**’ if:

- the system is in local mode or
- if the channel is ON (it is necessary to shut-down the channel with MOFF command first).

‘SIP’ Command

The **‘SIP’** command sets the IP address of the power supply A2630BS controller. The command has the form of ‘**SIP:*value*\r**’, where ***value*** is the IP address and is formed by four decimal numbers (each ranging from 0 to 255) separated by dots (for example: 'SIP:192.168.0.100\r'). The A2630BS controller will reply with a ‘**#NAK\r**’ if:

- command is incorrectly fed,
- channel is ON,
- the system is in local mode.

It is very important to notice that once the SIP command has been executed, the user can communicate and get control of the power supply again only by opening a new TCP socket with the just set IP.

‘MAC’ Command

The **‘MAC\r’** command returns the MAC and IP address of the power supply A2630BS controller in the form: **‘#MAC:*value1,value2*\r’**, where ***value1*** is the MAC address and the ***value2*** is the IP address of the channel.

‘BON’ Command

The **‘BON\r’** command turns the Bulk power supply on. Up to four channels can be connected to the same bulk unit: the bulk can be turned on by any of the four connected channels. Replies from the power supply A2630BS controller are in the form ‘**#AK\r**’ or ‘**#NAK\r**’ if the system is in local mode.

‘BOFF’ Command

The **‘BOFF\r’** command pulls down the line of bulk power but in order to turn the bulk OFF it is necessary to send the ‘BOFF\r’ command to all the power supplies connected to the bulk (up to four per crate). Replies from the power supply A2630BS controller are in the form ‘**#AK\r**’ or ‘**#NAK\r**’ if the system is in local mode.

‘VER’ Command

The **‘VER\r’** command returns the firmware version of power supply A2630BS controller. Reply from the A2630BS controller is ‘**VER:*value*\r**’, where ***value*** is the ID of firmware release.

EEPROM "Value" Cell Structure		
Cell #	**Cell caption**	**Protected**
0	c0 I_set	YES
1	c1 I_set	YES
2	c2 I_set	YES
3	c3 I_set	YES
4	I_max	YES
5	c0 V_read	YES
6	c1 V_read	YES
7	c2 V_read	YES
8	c3 V_read	YES
9	c0 DC_Link	YES
10	c1 DC_Link	YES
11	c2 DC_Link	YES
12	c3 DC_Link	YES
13	Kp (proportional constant)	NO
14	Ki (integral constant)	NO
15	Kd (derivative constant)	NO
18	Newton-Raphson Iterations	YES
19	Diagnostic Routine Iterations	YES
20	Max MOSFET Temperature	YES
21	Max SHUNT Temperature	YES
22	Serial Number	YES
23	Undervoltage Protection	YES
24 .. 25	*not used*	YES
26	Calibration Date	YES
27	Identification	NO
29	GP_IN Interlock Direction	YES
30	Slew Rate [A/s]	NO
31	Earth Current Limit [A]	NO
32	c0 Earth Current	YES
33	c1 Earth Current	YES
34 .. 37	*not used*	YES
38	Maximum current ripple threshold	NO
39 .. 45	*not used*	YES
46	Regulation Fault Enabled	NO
47	Warning Enabled	NO
48	Interlock Enable/Disable Mask	NO
49	Interlock Activation State Mask	NO
50	Interlock #0 Intervention Time (ms)	NO
51	Interlock #1 Intervention Time (ms)	NO
52	Interlock #2 Intervention Time (ms)	NO
53	Interlock #3 Intervention Time (ms)	NO
54	Interlock #4 Intervention Time (ms)	NO
55	Interlock #5 Intervention Time (ms)	NO
56	Interlock #6 Intervention Time (ms)	NO
57	Interlock #7 Intervention Time (ms)	NO
58	Interlock #8 Intervention Time (ms)	NO

Table 3: EEPROM Cells structure